

Strategic Business Requirements

MEDICAL RESEARCH AI ASSISTANT

From: Business Owner

To: Product Owner (PO)

Hi Team — Please use these core business requirements to map out our initial Agile product backlog, structure the high-level Epics, and draft the first set of User Stories. Our primary objective is to deliver a highly accurate, intuitive tool that medical professionals can fully trust for clinical document interrogation. Let's keep the focus on absolute data integrity and conversational simplicity.

1. PRODUCT VISION & COMMERCIAL PURPOSE

We want to build a straightforward, highly reliable Medical Research AI Assistant. The goal is to help our research teams cut through the noise by giving them a simple way to upload, search, and interrogate complex medical literature. By leveraging a combination of smart document retrieval and language models, the platform should deliver fast, conversational answers rooted strictly in the provided texts. The end product needs to balance ease of use with absolute factual transparency.

2. HIGH-LEVEL AGILE EPICS

To keep our development cycle moving efficiently, I have broken down our core requirements into five key business themes. Please use these to guide your sprint planning and backlog prioritization.

Epic 1: Effortless Ingestion & Paper Parsing

Business Value: Users should be able to upload large sets of research materials quickly without losing the underlying formatting or scientific layout that gives the data context.

- **Clean Upload Experience:** Provide a smooth frontend upload area featuring intuitive drag-and-drop or simple file selection tools.
- **Support for Key Formats:** Ensure the system natively parses and reads both PDF and XML medical research documents seamlessly.
- **Preserving Paper Structure:** The system must recognize standard scientific sections (like the Abstract, Methods, Results, and Discussion) so that text fragments maintain their structural context.

Epic 2: Medical Terminology Mapping & Intelligent Search

Business Value: General search tools don't work for clinical research. The system must understand complex medical vocabulary and easily navigate thousands of documents simultaneously.

- **Hybrid Search Architecture:** Combine deep semantic meaning-based search with strict, exact keyword matching so specialized vocabulary is never missed.
- **Medical Vocabulary Accuracy:** Optimize the search layer to accurately identify specific medical shorthand, abbreviations, drug names, and biomarkers.
- **Smart Filtering Controls:** Give users a clean interface sidebar to filter down their search by key details like publication year, specific journals, and study types.
- **Cross-Document Synthesis:** Allow users to run queries that pull and merge insights across thousands of uploaded documents at the same time.

Epic 3: Grounded Conversational Q&A

Business Value: Enable researchers to ask questions in plain English without risking factual errors. The system should act purely as an objective lens for the uploaded papers.

- **Natural Conversational Input:** Provide a highly responsive chat box that can handle complex, multi-layered research queries smoothly.
- **Isolated Context:** Restrict the AI to answer exclusively using information found within the user's uploaded documents. Out-of-context assumptions or external knowledge base access must be strictly blocked.

Epic 4: Full Traceability & Interactive Citations

Business Value: Medical teams cannot accept a "black box." Every claim the AI makes must be instantly verifiable against the source literature.

- **Clickable Citations:** Display clear, interactive citations directly alongside generated responses, showing the specific paper and section context.
- **Source Text Transparency:** Give users immediate access to view the exact, unedited source text block used to construct the answer through a clear interface viewer.

Epic 5: Operational Ingestion Logs & Compliance

Business Value: To meet strict institutional compliance, we need a complete, transparent audit trail of how the system is being utilized.

- **Comprehensive Audit Infrastructure:** Maintain clean historical logs of user questions, applied filters, retrieved context blocks, generated responses, and system latency times.

3. STRICT BUSINESS GUARDRAILS & SAFE REFUSAL MANDATES

Because this system handles critical medical information, the product backlog must enforce clear, non-negotiable boundaries to guarantee safety and integrity.

Mandate 1: Zero-Fabrication Rule

The system is strictly prohibited from inventing or guessing medical findings. If a statement cannot be explicitly proven by the uploaded text, the system must not attempt to answer.

Mandate 2: Automated Safe Refusals

Build automated confidence checks directly into the backend. The platform must provide a polite, standardized refusal message if there is a lack of clear evidence, low retrieval confidence, or irrelevant search results.

4. NON-FUNCTIONAL CRITERIA & OPERATIONAL BENCHMARKS

To ensure this tool is production-ready, please verify that the underlying architecture matches these primary business performance goals:

- **High-Volume Throughput:** The ingestion system must reliably process files when teams upload thousands of deep XML and PDF research papers simultaneously.
- **Low Latency:** Keep the entire data pipeline highly optimized to ensure rapid response times and a snappy conversational experience.
- **System Resilience:** Implement clean, graceful error handling for document parsing errors or API drops, paired with proactive system health checks.
- **Clean Professional UI:** The layout must be highly responsive, modern, and free of clutter to support focused professional workflows.